

Categories, Proofs, and Processes: Assignment 1

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In category theory, for a statement P , it is not necessary that $\neg\neg P \neq P$. Proof by contradiction is incorrect.

Question 1

For any computable functions f, g of L , where $\text{cod}(f) = \text{dom}(g)$, their composition $g \circ f$ is also computable and hence in L . The identity function can be defined for each data type A , where $\forall x \in A : id_A(x) = x$, for which the identity axiom holds. Since functions always produce an output, the associativity axiom will also hold. Thus, this is an associated category of L .

Mention assumptions given. Haskell is not a category.

By definition of composition for computable functions f, g of L ,

$$g \circ f = \lambda x. g(f(x))$$

The fact that the computable functions always return an output is necessary since

$$h \circ (g \circ f) = h \circ \lambda x. g(f(x)) = \lambda x'. h((\lambda x. g(f(x)))(x')) = \lambda x'. h(g(f(x')))$$

$$(h \circ g) \circ f = \lambda x. (h \circ g)(f(x))$$

If f is not computable and $h = c$ for some constant c , then we have $h \circ (g \circ f) = c$, and $h \circ (g \circ f)$ being uncomputable,

$$h \circ (g \circ f) \neq (h \circ g) \circ f$$

The associativity axiom does not hold.

A monoid is a one-object category, a groupoid is a one-object category where all morphisms are iso. A monoid homomorphism is precisely a functor between monoids as one-object categories.

Question 2

The collections $\text{Ob}(\mathcal{C})$, $\text{Ar}(\mathcal{C})$ are already defined, along with the domain and codomain as the sets in the binary relations.

Proposition 2.1. *Rel composition satisfies the associativity axiom.*

Proof. Let $Q : W \rightarrow X, R : X \rightarrow Y, S : Y \rightarrow Z$ be arbitrary relations, so $Q \subseteq W \times X, R \subseteq X \times Y, S \subseteq Y \times Z$. Then

$$\begin{aligned} S \circ (R \circ Q)(w, z) &\iff \exists y : (R \circ Q)(w, y) \wedge S(y, z) \\ &\iff \exists y : (\exists x : Q(w, x) \wedge R(x, y)) \wedge S(y, z) \\ &\iff \exists x, y : Q(w, x) \wedge R(x, y) \wedge S(y, z) \\ &\iff \exists x, y : Q(w, x) \wedge (R(x, y) \wedge S(y, z)) \\ &\iff \exists x : Q(w, x) \wedge (\exists y : R(x, y) \wedge S(y, z)) \\ &\iff \exists x : Q(w, x) \wedge (S \circ R)(x, z) \\ &\iff (S \circ R) \circ Q(w, z) \end{aligned} \tag{1}$$

□

Proposition 2.2. *Rel composition satisfies the identity axiom.*

Proof. Let $R : X \rightarrow Y$ be an arbitrary relation. By definition of the identity relation, for set Z ,

$$\text{id}_Z = \{(a, a) | a \in Z\}$$

so

$$R \circ \text{id}_X = R = \text{id}_Y \circ R$$

□

Expand relation for identity.

Question 3

Proposition 3.3. *In **Set**, an arrow is epic if and only if it is surjective.*

Proof. Suppose $f : X \rightarrow Y$ is surjective, and that $g \circ f = h \circ f$, where $g, h : Y \rightarrow Z$. By surjectivity,

$$\forall y \in Y, \exists x \in X : f(x) = y$$

Applying either g or h , we have

$$\forall y \in Y, \exists x \in X : g \circ f(x) = g(y)$$

$$\forall y \in Y, \exists x \in X : h \circ f(x) = h(y)$$

Since $g \circ f = h \circ f$,

$$\forall y \in Y, \exists x \in X : g \circ f(x) = g(y) = h \circ f(x) = h(y)$$

$$\forall y \in Y : g(y) = h(y)$$

so $g = h$. We have

$$\forall g, h : g \circ f = h \circ f \Rightarrow g = h$$

so f is epic.

Suppose f is not surjective. Then

$$\exists y \in Y, \forall x \in X : f(x) \neq y$$

Let $y' \in Y$ be such an element. Define $Z = \{0, 1\}$, where $g = 0$, and $\forall y \in Y - \{y'\}, h(y) = 0$, with $h(y') = 1$. Clearly, $g \neq h$ since $g(y') \neq h(y')$. Then we have

$$\forall x \in X : g \circ f(x) = h \circ f(x) = 0 \wedge f(x) \neq y'$$

but $g \neq h$. Thus, f is not epic.

□

Only use this, no negations, for $\chi_{\text{im}(f)} : Y \rightarrow \{0, 1\}$,

$$\forall y \in Y : \chi_{\text{im}(f)}(y) = 1 \iff \exists x \in X : f(x) = y$$

Question 4

(a)

Proposition 4.4. *If f, g are arrows such that $g \circ f$ is monic, then f is monic.*

Proof. Suppose $g \circ f$ is monic and for arbitrary arrows h', k' (where the codomain matches f),

$$f \circ h' = f \circ k'$$

Since $g \circ f$ is monic, we have

$$\forall h, k : g \circ f \circ h = g \circ f \circ k \implies h = k$$

Composing g to the first equation,

$$g \circ f \circ h' = g \circ f \circ k'$$

Using $g \circ f$ being monic, then $h' = k'$. Thus, with h', k' being arbitrary, we have that

$$\forall h, k : f \circ h = f \circ k \implies h = k$$

So f is monic. □

(b)

Proposition 4.5. *If f is split epic, then it is epic.*

Proof. Suppose f is split epic, and for arbitrary arrows h', k' (where the domain matches f),

$$h' \circ f = k' \circ f$$

Since f is split epic,

$$\exists g : g \circ f = \text{id}$$

Precomposing g to the first equation,

$$h' \circ f \circ g = k' \circ f \circ g$$

$$\implies h' \circ \text{id} = k' \circ \text{id}$$

$$\implies h' = k'$$

Since h', k' are arbitrary, we have that

$$\forall h, k : k \circ f = h \circ f \implies h = k$$

So f is epic. □

If $f : A \rightarrow B$ monic, then $h : \mathcal{C}(X, A) \rightarrow \mathcal{C}(X, B)$ is injective.

(c)

Proposition 4.6. *If f and $g \circ f$ are iso, then g is iso.*

Proof. Suppose f and $g \circ f$ are iso. Let $f : A \rightarrow B, g : B \rightarrow C$. Then

$$\exists h : h \circ f = \text{id}_A \wedge f \circ h = \text{id}_B$$

$$\exists h' : h' \circ g \circ f = \text{id}_A \wedge g \circ f \circ h' = \text{id}_C$$

Then we can show that $h' \circ g = h$, by composing $h' \circ g \circ f = \text{id}_A$ with h ,

$$h' \circ g \circ f \circ h = \text{id}_A \circ h$$

$$\implies h' \circ g \circ \text{id}_A = h$$

$$\implies h' \circ g = h$$

Define $h'' = f \circ h'$. Then we have

$$h'' \circ g = f \circ h' \circ g = f \circ h = \text{id}_B$$

$$g \circ h'' = g \circ f \circ h' = \text{id}_C$$

So g is iso. □

(d)

Proposition 4.7. *If f is monic and split epic, then it is iso.*

Proof. Let $f : A \rightarrow B$ be monic and split epic. Since f is split epic,

$$\exists k : f \circ k = \text{id}_B$$

Suppose $k \circ f \neq \text{id}_A$. Let $g = k \circ f, h = \text{id}_A$ so $g \neq h$. Then

$$f \circ g = f \circ k \circ f = \text{id}_B \circ f = f$$

$$f \circ h = f \circ \text{id}_A = f$$

So $f \circ g = f \circ h = f$ but $g \neq h$. Thus f is not monic, a contradiction. So we must have $k \circ f = \text{id}_A$. As $k \circ f = \text{id}_A$ and $f \circ k = \text{id}_B$, f is iso. □

Proof. Since $f : A \rightarrow B$ is split epic, suppose we have $g : B \rightarrow A$ such that $f \circ g = \text{id}_B$. Since f is monic,

$$f \circ g \circ f = f \circ \text{id}_A \implies g \circ f = \text{id}_A$$

So f is iso. □

(e)

Corollary 4.8. *If f and $f \circ g$ are iso, then g is iso.*

Corollary 4.9. *If f is epic and split monic, then it is iso.*

Question 5

(a)

Proposition 5.10. *i is a monomorphism and an epimorphism.*

Proof. Let $(G, \cdot, 1_G)$ be an arbitrary monoid where we have arrows $g, h : (G, \cdot, 1_G) \rightarrow (\mathbb{N}, +, 0)$, and $i \circ g = i \circ h$. Since i is an inclusion map, its underlying function is the identity on $\mathbb{N}$, $|i| = \text{id}_{\mathbb{N}}$. We have

$$\begin{aligned} \forall k \in G : |i \circ g|(k) &= |i \circ h|(k) \\ \implies \forall k \in G : |i| \circ |g|(k) &= |i| \circ |h|(k) \\ \implies \forall k \in G : |g|(k) &= |h|(k) \end{aligned}$$

Then $g = h$, so i is monic.

Similarly, let $(G', \cdot, 1_{G'})$ be an arbitrary monoid where we have arrows $g', h' : (\mathbb{Z}, +, 0) \rightarrow (G', \cdot, 1_{G'})$, and $g' \circ i = h' \circ i$. As before,

$$\begin{aligned} \forall n \in \mathbb{N} : |g' \circ i|(n) &= |h' \circ i|(n) \\ \implies \forall n \in \mathbb{N} : |g'| \circ |i|(n) &= |h'| \circ |i|(n) \\ \implies \forall n \in \mathbb{N} : |g|(n) &= |h|(n) \end{aligned} \tag{2}$$

To prove the equality for $n < 0$, we use the structure preserving nature of monoid homomorphisms for g', h' . By definition,

$$\begin{aligned} \forall n_1, n_2 \in \mathbb{Z} : |g'|(n_1 + n_2) &= |g'|(n_1) \cdot |g'|(n_2), \quad |g'|(0) = 1_{G'} \\ \forall n_1, n_2 \in \mathbb{Z} : |h'|(n_1 + n_2) &= |h'|(n_1) \cdot |h'|(n_2), \quad |h'|(0) = 1_{G'} \end{aligned}$$

Then we see that $|g'|(0) = |h'|(0) = 1_{G'}$, and if we set $n_2 = -n_1$, we have

$$\begin{aligned} \forall n \in \mathbb{Z} : |g'|(0) &= |g'|(n) \cdot |g'|(-n) = 1_{G'} \\ \forall n \in \mathbb{Z} : |h'|(0) &= |h'|(n) \cdot |h'|(-n) = 1_{G'} \end{aligned} \tag{3}$$

By symmetry, since $n \in \mathbb{Z}$, we also have

$$\begin{aligned} \forall n \in \mathbb{Z} : |g'|(0) &= |g'|(-n) \cdot |g'|(n) = 1_{G'} \\ \forall n \in \mathbb{Z} : |h'|(0) &= |h'|(-n) \cdot |h'|(n) = 1_{G'} \end{aligned} \tag{4}$$

If $n > 0$, then $|g|(n) = |h|(n)$ from (2), so using (3),

$$\forall n \in \mathbb{N} : |h'|(n) \cdot |g'|(-n) = 1_{G'}$$

Preoperating with $|h'|(-n)$,

$$\forall n \in \mathbb{N} : |h'|(-n) \cdot |h'|(-n) \cdot |g'|(-n) = |h'|(-n) \cdot 1_{G'}$$

$$\implies \forall n \in \mathbb{N} : 1_{G'} \cdot |g'|(-n) = |h'|(-n) \quad \text{by (4)}$$

$$\implies \forall n \in \mathbb{N} : |g'|(-n) = |h'|(-n)$$

Thus, we have

$$\forall n \in \mathbb{Z} : |g'|(-n) = |h'|(-n)$$

so $g' = h'$, and i is epic. □

Proof. After proving $\forall n \in \mathbb{N} : g(n) = h(n)$, suppose we have a monoid M and $g, h : (\mathbb{Z}, +, 0) \rightarrow M$. By monoid homomorphisms,

$$g(-n) = g(-1 + \dots + -1) = g(-1) \cdot \dots \cdot g(-1) \quad (n \text{ times})$$

Similarly,

$$h(-n) = h(-1 + \dots + -1) = h(-1) \cdot \dots \cdot h(-1) \quad (n \text{ times})$$

So we just have to prove $g(-1) = h(-1)$ to have $\forall n \in \mathbb{Z} : g(n) = h(n)$

$$\begin{aligned} h(-1) &= h(-1) \cdot 1_M \\ &= h(-1) \cdot g(0) \\ &= h(-1) \cdot g(1 - 1) \\ &= h(-1) \cdot g(1) \cdot g(-1) \\ &= h(-1) \cdot h(1) \cdot g(-1) \\ &= h(-1 + 1) \cdot g(-1) \\ &= h(0) \cdot g(-1) \\ &= 1_M \cdot g(-1) \\ &= g(-1) \end{aligned} \tag{5}$$

□

(b)

Corollary 5.11. *i is not an iso.*

Proof. Let $j : (\mathbb{Z}, +, 0) \rightarrow (\mathbb{N}, +, 0)$ be an arbitrary arrow. Since the underlying function of $i, |i|$, is not surjective, we will always have

$$|i| \circ |j| \neq \text{id}_{\mathbb{Z}}$$

so $i \circ j \neq \text{id}_{\mathbb{Z}}$. i is not an iso. □

Question 6

(a)

Proposition 6.12. *For all objects A, A', B, B' of a connected groupoid $\mathcal{G}$, there is a bijection $\mathcal{G}(A, B) \cong \mathcal{G}(A', B')$.*

Proof. Suppose we have arrows $k_1 : A' \rightarrow A, k_2 : B \rightarrow B'$, which exist by definition. Their unique inverses $k_1^{-1} : A \rightarrow A', k_2^{-1} : B' \rightarrow B$ also exist since the arrows are iso.

We can define a mapping between homsets $h : \mathcal{G}(A, B) \rightarrow \mathcal{G}(A', B')$ where

$$\forall i \in \mathcal{G}(A, B) : h(i) = k_2 \circ i \circ k_1$$

Suppose we choose arbitrary $i, j \in \mathcal{G}(A, B)$ such that $h(i) = h(j)$. Then

$$\begin{aligned} h(i) = h(j) &\implies k_2 \circ i \circ k_1 = k_2 \circ j \circ k_1 \\ &\implies k_2^{-1} \circ k_2 \circ i \circ k_1 \circ k_1^{-1} = k_2^{-1} \circ k_2 \circ j \circ k_1 \circ k_1^{-1} \\ &\implies \text{id}_B \circ i \circ \text{id}_A = \text{id}_B \circ j \circ \text{id}_A \\ &\implies i = j \end{aligned} \tag{6}$$

So h is injective.

$$\forall i, j \in \mathcal{G}(A, B) : h(i) = h(j) \implies i = j$$

To prove surjectivity, for each arrow $i' \in \mathcal{G}(A', B')$, define $i = k_2^{-1} \circ i' \circ k_1^{-1}$ so

$$h(i) = k_2 \circ k_2^{-1} \circ i' \circ k_1^{-1} \circ k_1 = \text{id}_{B'} \circ i' \circ \text{id}_{A'} = i'$$

Thus, h is bijective. All homsets have the same cardinality. $\square$

$$\mathcal{G}(A, B) \cong \mathcal{G}(A', B) \cong \mathcal{G}(A, B') \cong \mathcal{G}(A', B')$$

(b)

Example 6.13. Objects are permutations of lists of size n where the entries are anything. Arrows are the $n!$ possible permutations of mappings between objects A, B , corresponding to a mapping from the indices of $a \in A$ to the indices of the $b \in B$.

For example, if $n = 2$, we can have objects $A = \{(0, 1), (1, 0)\}, B = \{(2, 3), (3, 2)\}$, with 2 arrows $f, g : A \rightarrow B$,

$$\begin{aligned} f(0, 1) &= g(1, 0) = (2, 3) \\ f(1, 0) &= g(0, 1) = (3, 2) \end{aligned}$$

Question 7

0 is the only zero object. It is the initial object since for all objects $m \in \mathbb{N}$, there exists the unique arrow of the $0 \times m$ matrix, and is the terminal object since for all objects $n \in \mathbb{N}$, there exists the unique arrow of the $n \times 0$ matrix.